

2024 Bi H1 Q20

Section: Sustainability and Interdependence

Topic: Crop Protection

Question Summary:

The Bt toxin gene can be inserted into maize plants using recombinant DNA technology. The question asks what effect this genetic modification produces.

Worked Solution:

- The Bt toxin gene is taken from the bacterium *Bacillus thuringiensis*.
- When inserted into the maize genome, the plant produces the Bt protein (toxin) in its tissues.
- This protein is toxic to certain insect larvae that feed on the crop, giving the plant resistance to insect pests.
- The gene does not directly affect resistance to viruses, herbicides, or drought.

Therefore, Bt maize is genetically modified to resist insect pests.

Final Answer: **C** (The maize will be resistant to insect pests).

Revision Tips:

- Bt = *Bacillus thuringiensis*, source of insecticidal gene.
- GM crops can reduce pesticide use but raise ecological concerns (gene transfer, pest resistance).

- Crop protection also includes biological control and use of selective herbicides.